

[illegible]

			120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	524	525	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540	541	542	543	544	545	546	547	548	549	550	551	552	553	554	555	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570	571	572	573	574	575	576	577	578	579	580	581	582	583	584	585	586	587	588	589	590	591	592	593	594	595	596	597	598	599	600	601	602	603	604	605	606	607	608	609	610	611	612	613	614	615	616	617	618	619	620	621	622	623	624	625	6
--	--	--	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	---

			241			360
1 <i>Leptolyngbya boryana</i> dg5 NifD	100.0%	100.0%	RDWVIFR-TDKKKDCTD.DPTQYDVAIG		DYNIIGDD-WSSRLLEEGLRVVAQWSGDCITNEINTPTVKLNLYCYRS-NY-SRH-EETYGIPW-EYNF-FGGTQIAKS	
2 <i>Nostoc</i> sp. PCC 7120 NifD	96.6%	73.0%	RDWVIFE-YDKLKKENRILDFESPVDVALIG		DYNIIGDD-WSSRLLEEGLRVVAQWSGDCITNEINTPTVKLNLYCYRS-NY-QRSLEEDYGWWEFNF-FGGTQIAKS	
3 <i>Trichormus variabilis</i> ATCC 29413 nif cluster1 NifD	96.4%	74.0%	RDWVIFE-YDKLKKENRILDFESPVDVALIG		DYNIIGDD-WSSRLLEEGLRVVAQWSGDCITNEINTPTVKLNLYCYRS-NY-QRSLEEDYGWWEFNF-FGGTQIAKS	
4 <i>Trichormus variabilis</i> ATCC 29413 nif cluster2 NifD	99.4%	85.8%	RDWVIFK-ADKKKEETLGFEFSPYDVAIG		DYNIIGDD-WSSRLLEEGLRVVAQWSGDCITLHEMMLTPVKLNLYCYRS-NY-ARH-EETYGIPW-EYNF-FGGTQIAKS	
5 <i>Crocospaera subtropica</i> ATCC 51142 NifD	97.2%	79.9%	RDWVIFT-ADKENAEK--GFEGETPYDVAIG		DYNIIGDD-WSSRLLEEGLRVVAQWSGDCITLTKWATPNVKLNLYCYRS-NY-SRH-EETYGIPW-EYNF-FGGTQIAKS	
6 <i>Synechococcus</i> sp. JA-3-3Ab NifD	98.6%	77.3%	RDWVIFI-ADKKAKKEELNLFESPVDVALIG		DYNIIGDD-WSSRLLEEGLRVVAQWSGDCITNEMLLAPRKNLYCYRS-NY-GRH-EETYGIPW-EYNF-FGGTQIAKS	
7 <i>Hydrococcus rivularis</i> NIES-593 nif cluster1 NifD	96.8%	83.0%	RDWVIFQ-ADKEN---EFESIPYDVAIG		DYNIIGDD-WSSRLLEEGLRVVAQWSGDCITNEMLLTPVKLNLYCYRS-NY-SRH-EETYGIPW-EYNF-FGGTQIAKS	
8 <i>Hydrococcus rivularis</i> NIES-593 nif cluster2 NifD	98.4%	37.7%	FKHWIGDTTETEEI-----EGFTVNLMG		EYNIIGDD-WEIERVLKCKGCTKATFSGDGYDAATKAHLAKVLYCHRSINYMATM-ETKYGIPWIKVNF-IGTHAFKSI	
9 <i>Pleurocapsales cyanobacterium</i> LEGE 06147 NifD	97.8%	37.7%	FKHWIGDTTETEEI-----EGFTVNLMG		EYNIIGDD-WEIERVLKCKGCTKATFSGDGYDAATKAHLAKVLYCHRSINYMATM-ETKFGVPIKVLNF-IGTHAFKSI	
10 <i>Phormidium</i> sp. PBR-2020 NifD	98.2%	37.5%	MKNVGTDEGAEEV-----EGYVNLMG		EYNIIGDD-WEIERVLKCKGCTKATFSGDGYDAATKAHLAKVLYCHRSINYVSEK-ETKFGTPIKVLNF-IGVNSFQAL	
11 <i>Trichormus variabilis</i> ATCC 29413 VnfDG	92.3%	25.1%	INEKVG-LEPEIT-----SPYINVLG		DYNIIGDDTFVLRYEYKMGQIIAHFTNGTYDRLGMHRAQLVNTNCARSAGY-IANELKKRYGIPRMDVDT-WGFDYCEAL	
12 <i>Nostoc carneum</i> NIES-2107 VnfDG1	92.3%	25.1%	INEKVG-LEPEIT-----SPYINVLG		DYNIIGDDTFVLRYEYKMGQIIAHFTNGTYDRLGMHRAQLVNTNCARSAGY-IANELKKRYGIPRMDVDT-WGFDYCEAL	
13 <i>Nostoc carneum</i> NIES-2107 VnfDG2	92.5%	22.2%	INEKVG-LEPEIT-----SPYINVLG		DYNIIGDDTFVLRYEYKMGQIIAHFTNGTYDRLGMHRAQLVNTNCARSAGY-IANELKKRYGIPRMDVDT-WGFDYCEAL	
14 <i>Hydrococcus rivularis</i> NIES-593 nif cluster2 NifE	89.7%	27.8%	FR-LVGT-GDTTGI-----SPHSINILG		DFNLAGEIMIRIYEFERMGTQVYANITGDRVGDIAHRAHGALNVVCSGATMDLAKMKKEGYIPFVRVSY-FGLDMADAL	
15 <i>Roseiflexus</i> sp. RS-1 NifD	90.9%	34.9%	FRQLVGS-VEPEGD-----FSRSINILG		DYNIKNDIRTFEYIPEALGLRIIRARFTGNVSDDLKIMHKALNLYHCORSATY-IADMMKDKGYIPYINVTL-WGMKNMAKAL	
16 <i>Phormidium</i> sp. PBR-2020 NifE	89.9%	28.6%	FR-LVGT-GDTSDI-----SPHSINILG		DFNLAGEIMIRIYEFERMGTQVYANITGDRVADISRSHGALNVVCSGATLDLAKMKKEGYIPFVRVSY-FGLDMADAL	
17 <i>Pleurocapsales cyanobacterium</i> LEGE 06147 NifE	89.5%	28.6%	FR-LVGT-GDTTGI-----SPHSINILG		DFNLAGEIMIRIYEFERMGTQVYANITGDRVGDIAHRAHGALNVVCSGATMDLAKMKKEGYIPFVRVSY-FGLDMADAL	
18 <i>Leptolyngbya boryana</i> dg5 NifE	91.5%	29.2%	LDYVVG-GEPEYT-----TPYDINILG		EYNIAGEMGVLPLEKLGIRVLAKITGDARYEITAYAHRAKNVIMCISKALINMGKKEERYGIPYIEESF-YGIDAMNRCI	
19 <i>Nostoc</i> sp. PCC 7120 NifE	91.7%	27.8%	LDYVVG-GEPEYT-----TPYDINILG		EYNIAGEMGVLPLEKLGIRVLSKITGDARFEITAYAHRAKNVIMCISKALLNMARKKEERYGIPYIEESF-YGIDAMNRCI	
20 <i>Trichormus variabilis</i> ATCC29413 nif cluster1 NifE	91.7%	27.4%	LDYVVG-GEPEYT-----TPYDINILG		EYNIAGEMGVLPLEKLGIRVLSKITGDARFEITAYAHRAKNVIMCISKALLNMARKKEERYGIPYIEESF-YGIDAMNRCI	
21 <i>Crocospaera subtropica</i> ATCC 51142 NifE	91.7%	27.3%	LDHVVG-KEPEYT-----TAYDINILG		EYNVAGEMGVLPLEKKGIRVLAKITGDARYEIVCSAHRAKNVIMCISKALINMASAKERYIPYIEESF-YGVQDMNRCI	
22 <i>Synechococcus</i> sp. JA-3-3Ab NifE	91.9%	26.8%	LDYVVG-ASPEYT-----TPYDINILG		EYNIAGELMDVLPLEDELGIRVLAKITGDARYEITCAHRAKNVILCISAKALLNVRGKKEERYGIPYIEESF-YGLQEMNHCI	
23 <i>Hydrococcus rivularis</i> NIES-593 nif cluster1 NifE	91.7%	26.1%	LEVVG-KEPEYT-----TPYDINILG		EYNIAGEMVNTPIFEKLGIRVLAKITGDARYEITCAHRAKNVIMCISKALINMATKMKKGYIPYIEESF-YGIDAMNRCI	
24 <i>Trichormus variabilis</i> ATCC29413 VnfE1	88.9%	19.0%	VDVMEPTPERQAQ-----LADSVLVGAKMGPTEVEGDKELKRLIKGMGARVHSVIAGDCTLEELQKASAAVNTCLDLGYITGSKAKSDRFETPLNSTILPYGISATEKAL		DYNIAGEMVNLPLLEKLGIRVLSKITGDARYEITCAHRAKNVITCSNVSLSKIAQKKEGYIPYIEESF-YGVENMSRCI	
25 <i>Nostoc carneum</i> NIES-2107 VnfE1	88.9%	18.6%	VDVMEPTPERKQS-----LADSVLVGAKMGPTEVEGDKELKRLIKGMGARVHSVIAGDCTLEELQKASAAVNTCLDLGYITGSKAKSDRFETPLNSTILPYGISATEKAL		DYNIAGEMVNLPLLEKLGIRVLSKITGDARYEITCAHRAKNVITCSNVSLSKIAQKKEGYIPYIEESF-YGVENMSRCI	
26 <i>Nostoc carneum</i> NIES-2107 NifE2/VnfE2	91.7%	26.3%	LDYVVG-ASPEFT-----TPYDINILG		DYNIAGEMVNLPLLEKLGIRVLSKITGDARYEITCAHRAKNVITCSNVSLSKIAQKKEGYIPYIEESF-YGVENMSRCI	
27 <i>Trichormus variabilis</i> ATCC29413 NifE4/VnfE2	91.5%	28.6%	LDYVVG-ASPEFT-----TPYDINILG		DYNIAGEMVNLPLLEKLGIRVLSKITGDARYEITCAHRAKNVITCSNVSLSKIAQKKEGYIPYIEESF-YGVENMSRCI	
consensus/100%	hp.h.s.-.t.....slslhg		hplts-hh...hhpthghphs.hsgssphtht.....stlshh.C.t.t...hsp.h.p.ashp.h.p.h.g.t.htt.1			
consensus 90%	hp.hhs...p.t.....shslslhg		-anltg-hh...hhcthgpllluphghssphphth.h...stlshh.C.tsh...hsp.hppaahp.h.ph.sh.g.p.htts			
consensus 80%	hc.hhsp.h.-p.h.....psaslsllg		-Ynlgt-ha...hh-chglcluphghpuph.ht.s.t.p.hh.C.cuh...huph.pccpplanc.sh.ag.ptshcs			
consensus 70%	hc.llss.s.-.chp.....cpaslnltg		-Ynlug-hhslc.lhcgcl+lhphghpuph.p.h.s.t.p.hh.C.+uh...luptch-cyglplanchsa.ahpohs			

			consensus 100%		361		I		480	
1	<i>Leptolyngbya boryana</i> dg5 NifD	100.0%	100.0%		REIAAKFDE	-----T-QAKAEVIAKYQ-OTDEVIAKFL-R-OGKTVALMVVGL-R-RHVVPAFY-DLGNRLIGTYEFCNDYDK-----				
2	<i>Nostoc</i> sp. PCC 7120 NifD	96.6%	73.0%		REIAAKFDS	-----KIQENAEVIAKYTPVMNVALDKYR-R-LEGNTVMLYVYGL-R-RHVVPAFE-DLGKVVGTGYEFAHNDYDK-----				
3	<i>Trichormus variabilis</i> ATCC 29413 nif cluster1 NifD	96.4%	74.0%		REIAAKFDS	-----KIQENAEVIAKYTPVMNVALDKYR-R-LEGNTVMLYVYGL-R-RHVVPAFE-DLGKVVGTGYEFAHNDYDK-----				
4	<i>Trichormus variabilis</i> ATCC 29413 nif cluster2 NifD	99.4%	85.8%		REIAAKFDE	-----T-QAKTEEVIAKYFA-OTKALVDKYRS-LEGKTVALMVVGL-R-RHVVPAFE-DLGKRLIGTYEFAHNDYDK-----				
5	<i>Crocospaera subtropica</i> ATCC 51142 NifD	97.2%	79.9%		REIAASFDE	-----KIQAKAEVIEIKYRKQSEELIAKYR-R-LEGKTVMVMVGL-R-RHVVPAFK-DLGNELIGTYEFAHNDYDK-----				
6	<i>Synechococcus</i> sp. JA-3-3Ab NifD	98.6%	77.3%		RAIAQRFDE	-----T-QAKAEVIAKYQOSEANLIAKYR-R-LEGKRVLMVVGGL-R-RHVIPAFR-ELGNEVLTGYEFCDDYDK-----				
7	<i>Hydrococcus rivularis</i> NIES-593 nif cluster1 NifD	96.8%	83.0%		RAIANFFDD	-----KIKKEGAEVIAKYQOQ-DEILAKYR-R-LEGKVMVMVVGGL-R-RHVVPAFK-DLGNEMLTGYEFCNDYDK-----				
8	<i>Hydrococcus rivularis</i> NIES-593 nif cluster2 NifD	98.4%	37.7%		RKIAARFFDN	-----PELTARVEELTTEETALAEAAQAEYRKR-LEGKTVELFVGGG-RAHHYQELFA-DLGNKTIIVAGYEFARDDYEGRQALPKIRVDADSRNIEELEVEPPERYK-----				
9	<i>Pleurocapsales cyanobacterium</i> LEGE 06147 NifD	97.8%	37.7%		RKIAARFFDN	-----PDILTARIEAVLAEVIALAEAAQAEYRKR-LEGKTVELFVGGG-RAHHYQELFA-DLGNKTIIVAGYEFARDDYEGRQALPKIKVDADSRNIEELEIEPDSERYK-----				
10	<i>Phormidium</i> sp. PBR-2020 NifD	98.2%	37.5%		RKIAEVFFDD	-----PKLTERVEIVIAEEMAE-TEAQLTYRERLTGKYVFLFVGGG-RAHHYQGLFA-DLGNKTIIVAGYEFARDDYEGRDLINPLNKIDADSRNIEEIHVKQDPERYK-----				
11	<i>Trichormus variabilis</i> ATCC 29413 VnfDG	92.3%	25.1%		RKIGAFHFG	-----TEDRAEAVIAEETIAKYQ-EMDWMYKERLGGKVCIVITGGP-RLWHNTKALEDDLGNQVVMSSSKFCHQEDFE-----				
12	<i>Nostoc carneum</i> NIES-2107 VnfDG1	92.3%	25.0%		RKIGAFHFG	-----TEDRAEAVIAEAAVAYQDRMMWYKERLGGKVCIVITGGP-RLWHNTKALEDDLGNQVVMSSSKFCHQEDFE-----				
13	<i>Nostoc carneum</i> NIES-2107 VnfDG2	92.5%	22.2%		RKIGAFHFG	-----TEDRAEAVIAEAAVAYQDRMMWYKERLGGKVCIVITGGP-RLWHNTKALEDDLGNQVVMSSSKFCHQEDFE-----				
14	<i>Hydrococcus rivularis</i> NIES-593 nif cluster2 NifE	89.7%	27.8%		YKVAHFHFG	-----DPEILKRAQVVRDEITAVLPKLGQYRKD-OGKKAITYVGGAFKAFSLVKAFR-LLGDMVMVVSQGTCTSDYR-----				
15	<i>Roselluxes</i> sp. RS-1 NifD	90.9%	34.9%		RDTAAFFDG	-----LEARAEEVIAREAVITQPYDAYROR-OGKRVFIYGGGP-RWHVIELLR-ELGNETATAITTECHDYDE-----				
16	<i>Phormidium</i> sp. PBR-2020 NifE	89.9%	28.6%		YKVAHFHFG	-----EPEILKRAQVVRDEITAVLPKLGQYRKD-OGKKAITYVGGAFKAFSLVKAFR-LLGDMVMVVSQGTCTSDYR-----				
17	<i>Pleurocapsales cyanobacterium</i> LEGE 06147 NifE	89.5%	28.6%		YKVAHFHFG	-----DSEILHRAQVVRDEITAVLPKLGQYRKD-OGKKAITYVGGAFKAFSLVKAFR-LLGDMVMVVSQGTCTSDYR-----				
18	<i>Leptolyngbya boryana</i> dg5 NifE	91.5%	29.2%		RNTAAKIGD	-----ANLQERVELLTADETAKLDEALAPYRARLKGKRVVLYTGGV-KSWSIISAQ-DLGNEMVATSTKSTEEKS-----				
19	<i>Nostoc</i> sp. PCC 7120 NifE	91.7%	27.8%		RNTAAKIGD	-----PDLOARTKLTAEETAALDLALAPYRARLKGKRVVLYTGGV-KSWSIISAQ-DLGNEMVATSTKSTEEKS-----				
20	<i>Trichormus variabilis</i> ATCC29413 nif cluster1 NifE	91.7%	27.4%		RNTAAKIGD	-----PDLOQRTKLTAEETAALDLALAPYRARLKGKRVVLYTGGV-KSWSIISAQ-DLGNEMVATSTKSTEEKS-----				
21	<i>Crocospaera subtropica</i> ATCC 51142 NifE	91.7%	27.3%		RNTAQHFGD	-----QVLIQERVEETIEEETAKLDEALAPYRERLKGKRVVLYTGGV-KSWSIISAQ-DLGNEMVATSTKSTEEKG-----				
22	<i>Synechococcus</i> sp. JA-3-3Ab NifE	91.9%	26.8%		RNTAQGLKD	-----PELQERTALITERETAEALAPYRQLAGKRVVLYTGGV-KSWSIISAAR-DLGNQVATLSTKSTEEKS-----				
23	<i>Hydrococcus rivularis</i> NIES-593 nif cluster1 NifE	91.7%	26.1%		RNAAKLLGD	-----ADLQERVELLTADETAKLDEALAPYRERLKGKRVVLYTGGV-KSWSIISAQ-DLGNEMVATSTKSTEEKS-----				
24	<i>Trichormus variabilis</i> ATCC29413 VnfE1	88.9%	19.0%		RGAAYKLYK	-----MEACABALMEREYAATKTEFAEAAKYIEGKLATIEGHDAIKCLSTIAHMLERDFGRRAVIYNFHPWSTEAR-----				
25	<i>Nostoc carneum</i> NIES-2107 VnfE1	88.9%	18.6%		QGAAYKLYG	-----MEACABALMEREYAATKTEFAEAAKYIEGKLATIEGHDAIKCLSTIAHMLERDFGRRAVIYNFHPWSTEAR-----				
26	<i>Nostoc carneum</i> NIES-2107 NifE2/VnfE2	91.7%	26.3%		RNTAAALGLD	-----PELQERTALITERETAEALAPYRQLAGKRVVLYTGGV-KSWSIISAAR-DLGNEMVATSTKSTEEKS-----				
27	<i>Trichormus variabilis</i> ATCC29413 NifE4/VnfE2	91.5%	28.6%		RNTAAALFGD	-----QYLQERTETILTAEETAALDLALAPYRSQ-OGKRVLYSGGV-KSWSIISAAR-DLGNEMVATSTKSTEEKS-----				

[illegible]